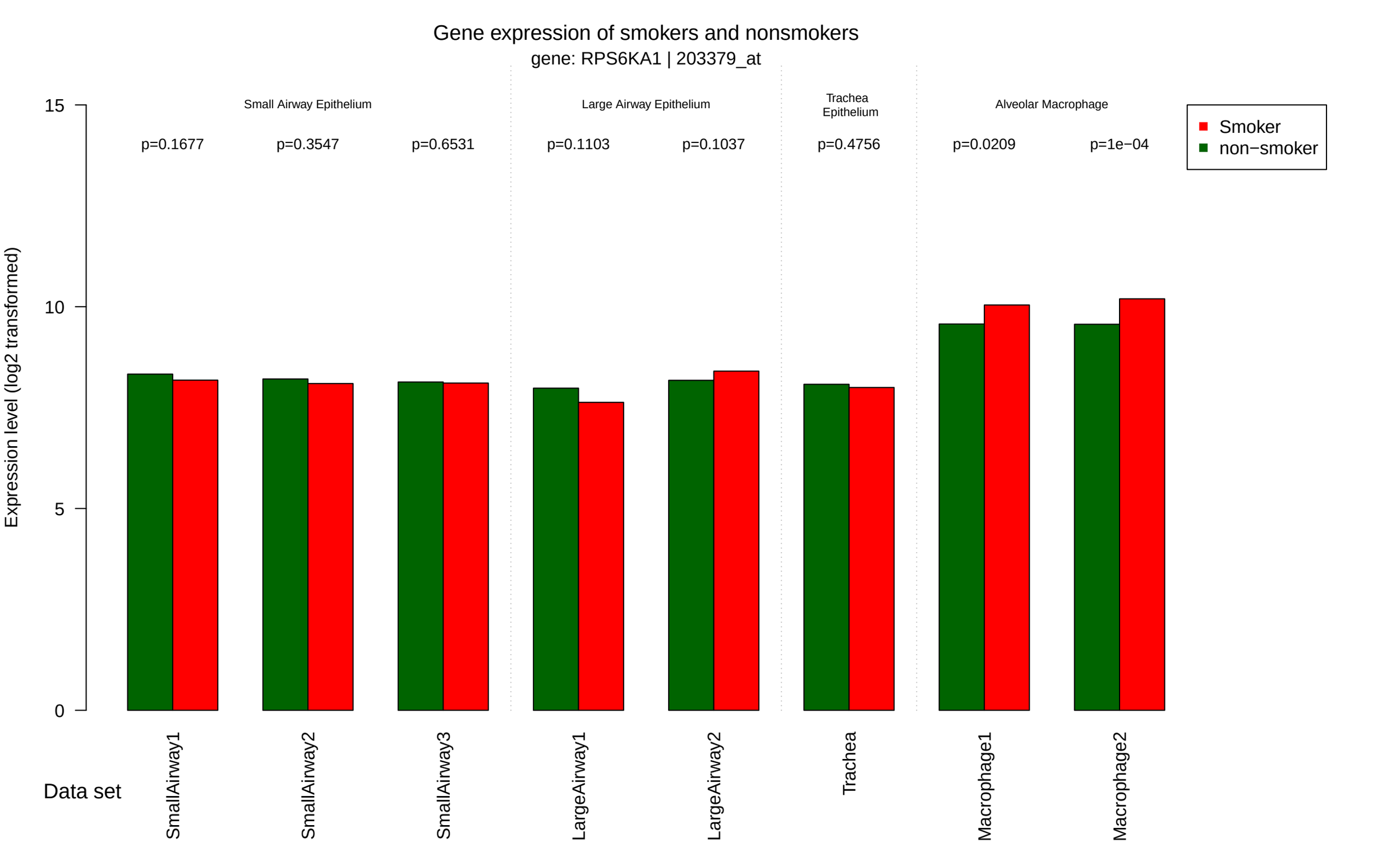
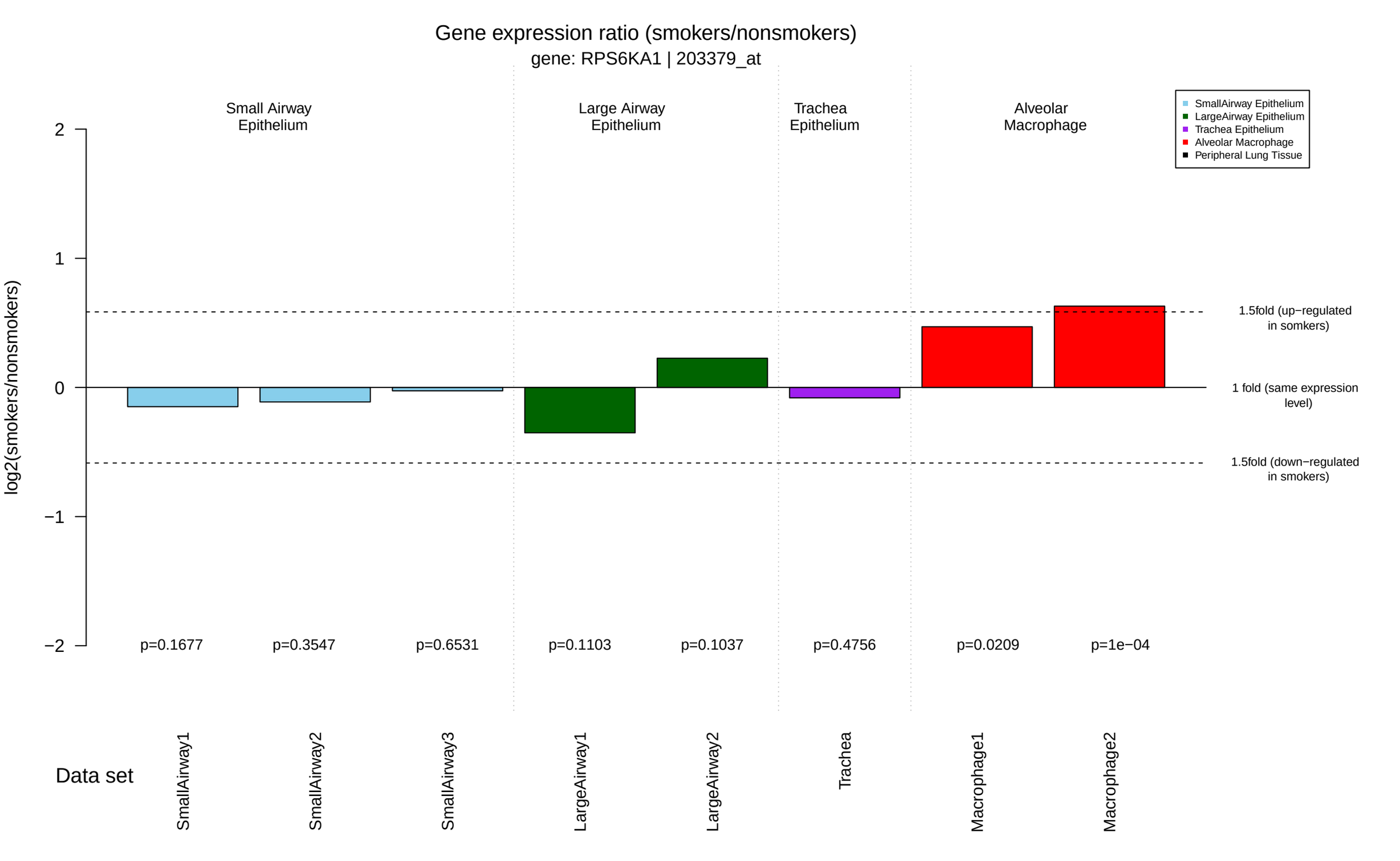
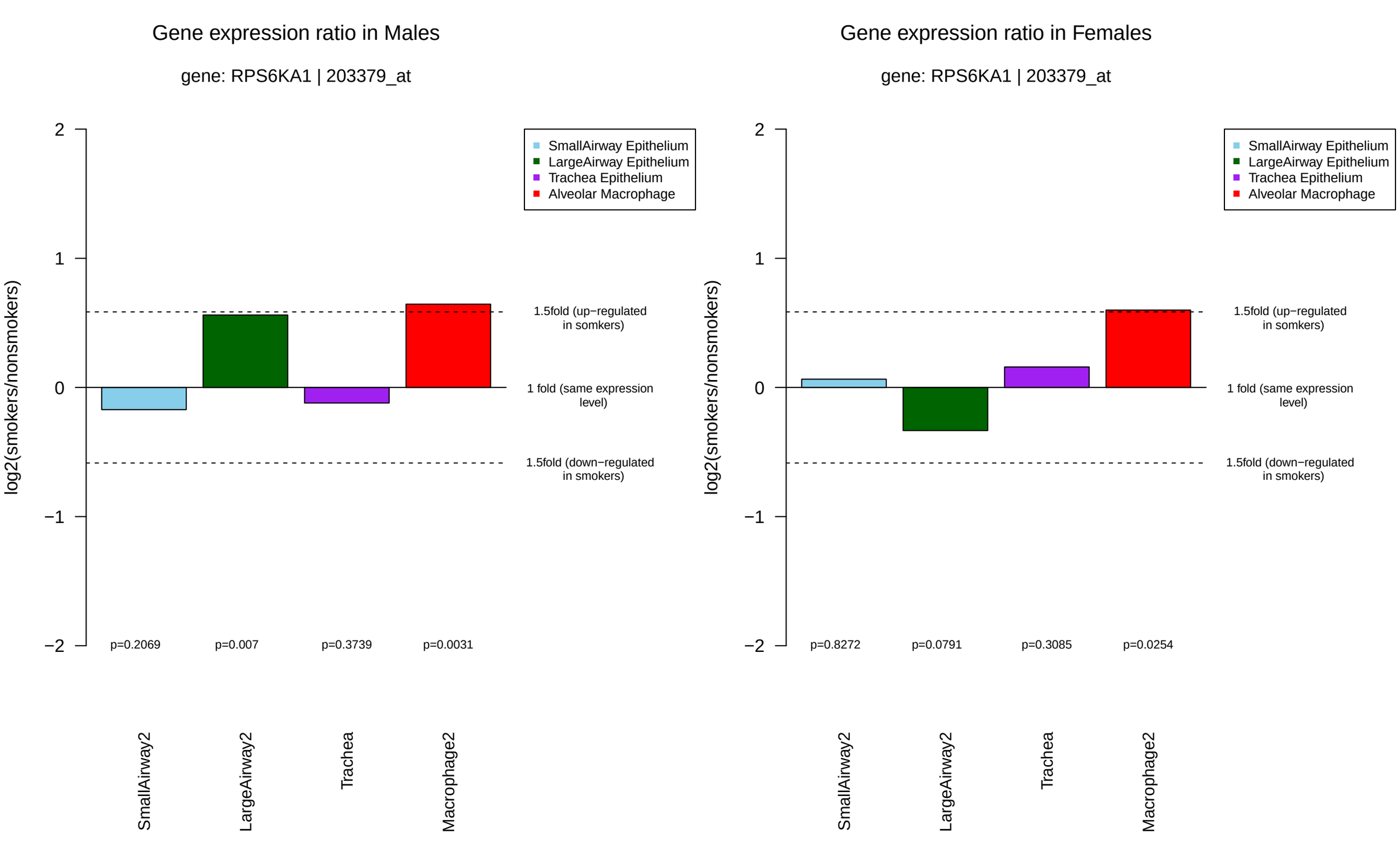
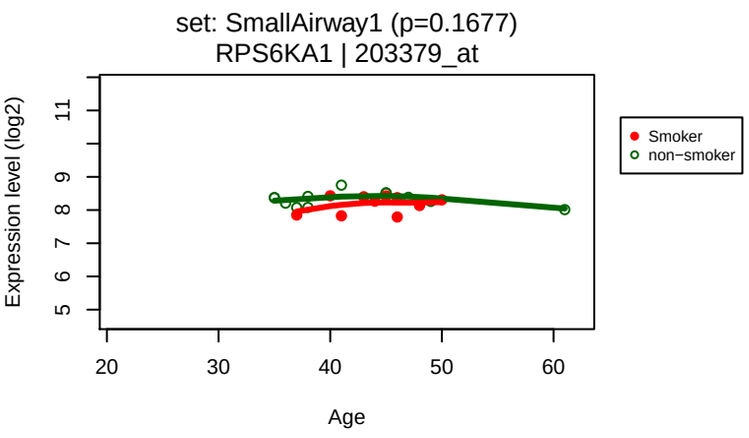
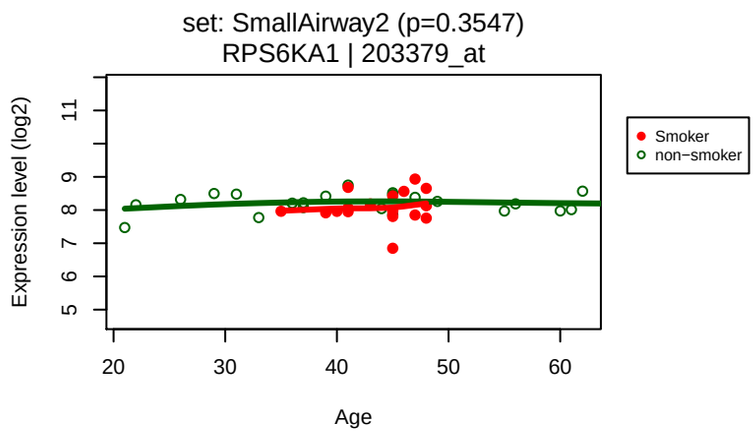
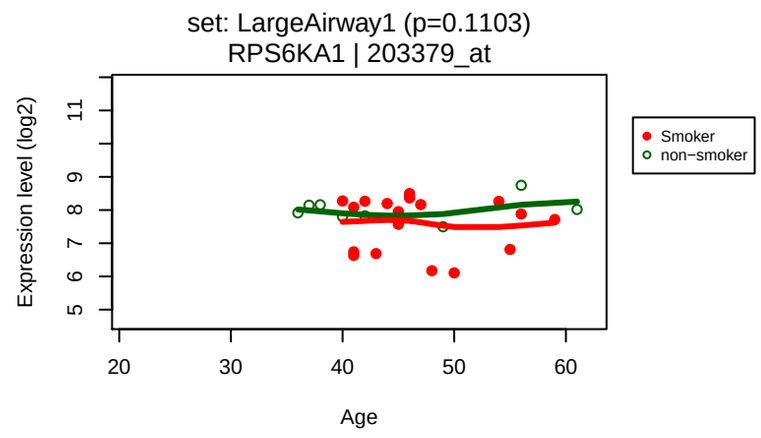

Gene expression of smokers and nonsmokers
gene: RPS6KA1 | 203379_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1677
p=0.3547
p=0.6531
p=0.1103
p=0.1037
p=0.4756
p=0.0209
p=1e−04
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

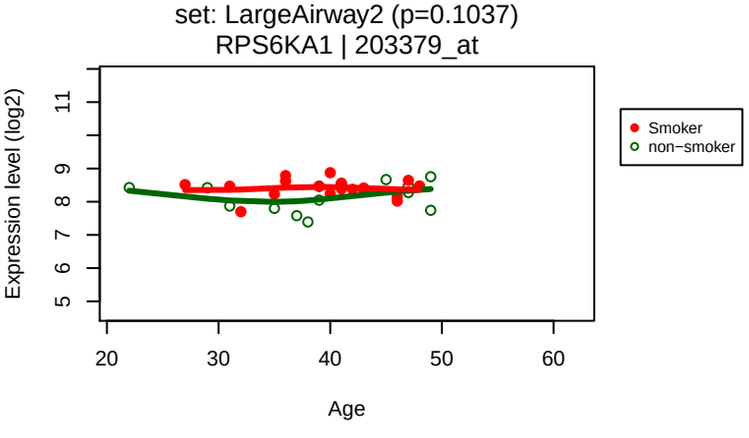

Gene expression ratio (smokers/nonsmokers)
gene: RPS6KA1 | 203379_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1677
p=0.3547
p=0.6531
p=0.1103
p=0.1037
p=0.4756
p=0.0209
p=1e−04
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

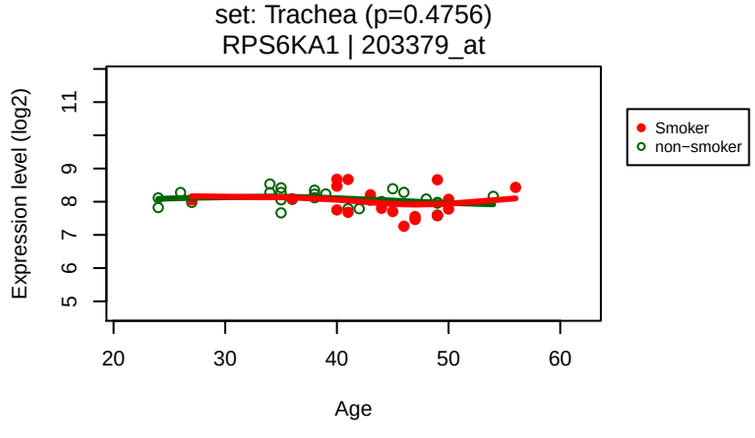

Gene expression ratio in Males
gene: RPS6KA1 | 203379_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2069
p=0.007
p=0.3739
p=0.0031
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPS6KA1 | 203379_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8272
p=0.0791
p=0.3085
p=0.0254
SmallAirway2
LargeAirway2
Trachea
Macrophage2

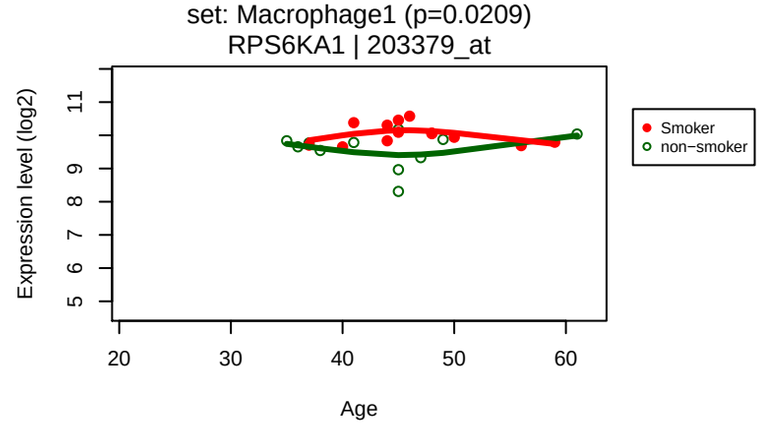

set: SmallAirway1 (p=0.1677)
RPS6KA1 | 203379_at
Expression level (log2)
Age
Smoker
non−smoker

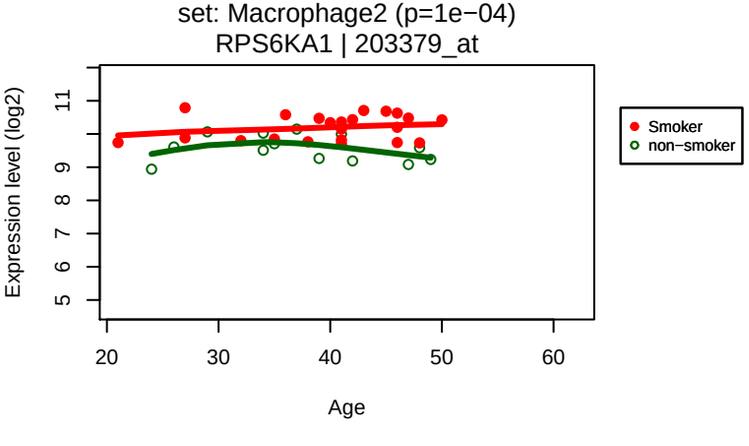

set: SmallAirway2 (p=0.3547)
RPS6KA1 | 203379_at
Expression level (log2)
Age
Smoker
non−smoker

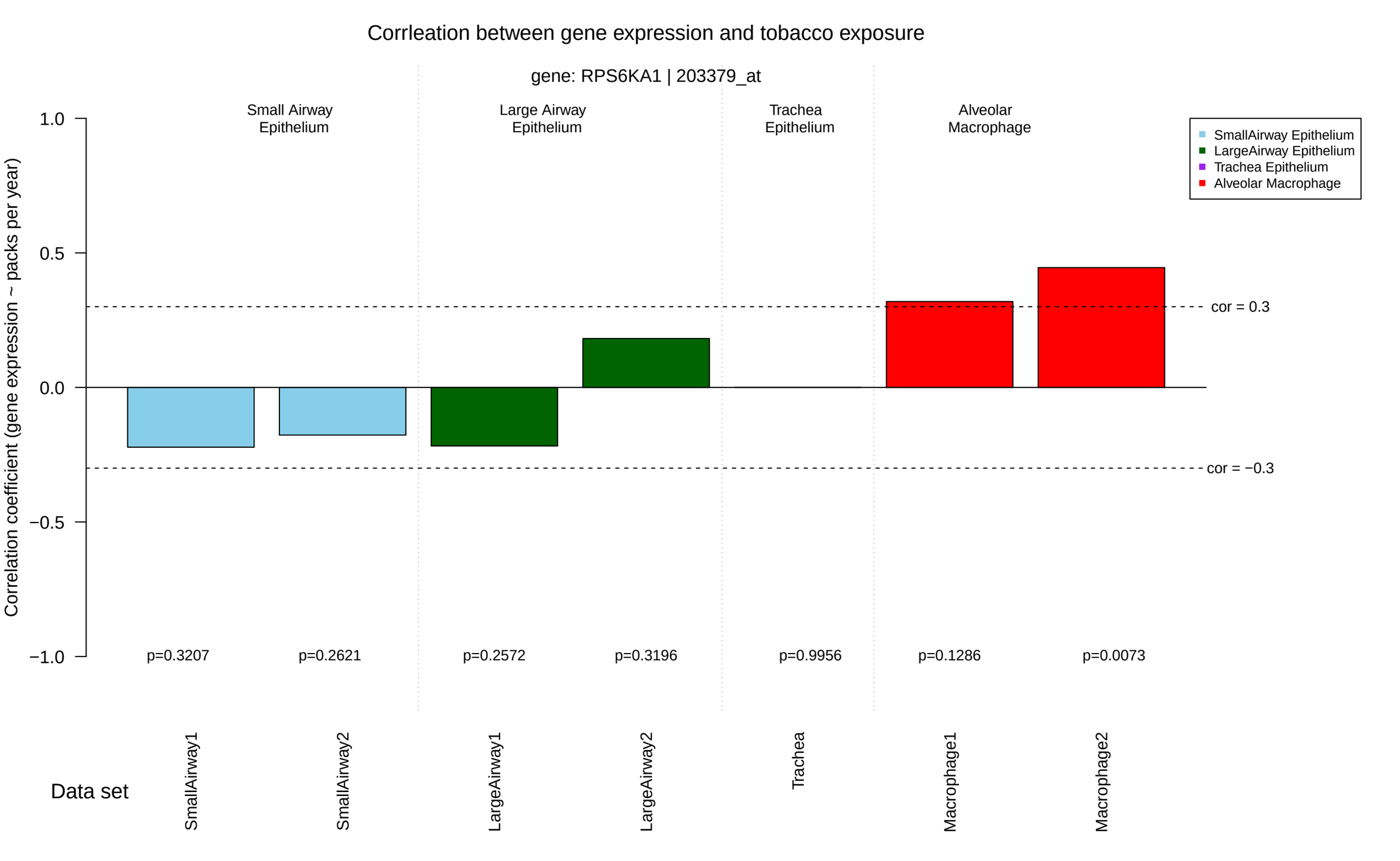

set: LargeAirway1 (p=0.1103)
RPS6KA1 | 203379_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1037)
RPS6KA1 | 203379_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4756)
RPS6KA1 | 203379_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0209)
RPS6KA1 | 203379_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=1e−04)
RPS6KA1 | 203379_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPS6KA1 | 203379_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3207
p=0.2621
p=0.2572
p=0.3196
p=0.9956
p=0.1286
p=0.0073
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2